



Assessment Analysis Guide
25-26 Math G5 Form L Mult/Div Mult-D #s
Mathematics
Grade 5

Reference #: 6851b4a

1

Item Id: i185090

Item Type: SR (Selected Response)

Standards Description:

5.NBT.B.5 Fluently multiply multi-digit whole numbers using the standard algorithm.

Content:

Solve.

$$1,469 \times 374 = ?$$

- | | |
|---|---------|
| A | 20,566 |
| B | 411,576 |
| C | 547,306 |
| D | 549,406 |

Primary Standard: 5.NBT.B.5

Points Possible: 1

Distractor Rationale:

A: 20,566

Student multiplied by single-digit values instead of correctly accounting for the place value of each digit when finding partial products.

B: 411,576

Student did not regroup when multiplying, but regrouped when adding.

C: 547,306

Student correctly found each partial product, but did not regroup when adding the partial products.

D: 549,406

Correct.

2

Item Id: i185092

Item Type: SR (Selected Response)

Standards Description:

5.NBT.B.6 Find whole-number quotients of whole numbers with up to four-digit dividends and two-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.

Content:

Divide.

$$2,752 \div 16 = ?$$

A

103

B

172

C

1,072

D

1,720

Primary Standard: 5.NBT.B.6**Points Possible: 1**

Distractor Rationale:

A: 103

Student incorrectly used partial quotients by combining the results as separate digits to form the final answer, rather than adding them, and ignored the remainder.

B: 172

Correct.

C: 1,072

Student miscalculated the product of the divisor and the partial quotient, leading to an incorrect initial estimate in the division process.

D: 1,720

Student incorrectly determined the number of digits in the quotient, assuming it should match the number of digits in the dividend.

3

Item Id: i185437

Item Type: SR (Selected Response)

Standards Description:

5.NBT.B.5 Fluently multiply multi-digit whole numbers using the standard algorithm.

Content:

What is the product?

$$729 \times 63 = ?$$

- | | |
|---|--------|
| A | 6,561 |
| B | 44,407 |
| C | 45,927 |
| D | 68,667 |

Primary Standard: 5.NBT.B.5

Points Possible: 1

Distractor Rationale:

A:	6,561	Student did not account for place value while multiplying the first factor by the value in the tens place of the second factor while performing the standard algorithm for multiplication.
B:	44,407	Student did not account for the "carried" digits while performing the standard algorithm for multiplication.
C:	45,927	Correct.
D:	68,667	Student added the "carried" digits before multiplying rather than after while performing the standard algorithm for multiplication.

4

Item Id: i185093

Item Type: SR (Selected Response)

Standards Description:

5.NBT.B.6 Find whole-number quotients of whole numbers with up to four-digit dividends and two-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.

Content:

Divide.

$$608 \div 19 = ?$$

A 30

B 32

C 38

D 50

Primary Standard: 5.NBT.B.6**Points Possible:** 1

Distractor Rationale:

A: 30

Student rounded the dividend to the nearest hundred and the divisor to the nearest ten then divided.

B: 32**Correct.**

C: 38

Student interpreted the amount remaining after removing multiples of 10 through partial quotients as the quotient.

D: 50

Student left-aligned the partial quotients, resulting in a place value error when adding them together.

5

Item Id: i185089

Item Type: SR (Selected Response)

Standards Description:

5.NBT.B.5 Fluently multiply multi-digit whole numbers using the standard algorithm.

Content:

Multiply.

$$689 \times 716 = ?$$

- A 9, 646
- B 59, 254
- C 436, 874
- D 493, 324

Primary Standard: 5.NBT.B.5**Points Possible:** 1

Distractor Rationale:

A:	9, 646	Student multiplied by single-digit values instead of correctly accounting for the place value of each digit when finding partial products.
B:	59, 254	Student miscalculated the place value and multiplied by ten times less than the correct value when finding partial products.
C:	436, 874	Student did not regroup when multiplying, but did regroup when adding partial products.
D:	493, 324	Correct.